SOFTWARE-DEFINED NETWORKING: A Revolution

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The research for a new paradigm for a better architecture of networking began in 2004. After several proposals and researches, the concept of software-defined networking (SDN) emerged in 2008. In 2011, Open Networking Foundation was formed to take this concept to an executable level. The foundation consists of 69 members including Cisco, Juniper, HP, Dell and IBM.

Google, Yahoo, DT, Microsoft and Facebook serve as the Board of this foundation. One of the first SDN projects was AT&T's GeoPlex. Some of the available commodity switches that are compliant with OpenFlow protocol of SDN are Hewlett-Packard (8200zl, 6600, 6200zl), Brocade (5400zl, 3500/3500yl) and IBM NetIron CES 2000 Series.

In traditional networks, most of the network functionality is implemented in hardware devices like switches and routers. In such systems, each appliance needs to be configured individually. Moreover, a router will treat all incoming packets equally and will follow the same protocols for all.

Inculcating such functionality is under the control of the device provider. So when a new technology, feature, update or application is introduced, all hardware devices need to be replaced. Thus, it has been found that traditional networking evolves slowly, is not dynamic and has limited functionality, as provided by the vendors. Being hardware-centric, it proves to be a huge challenge when the network is big and complex.

SDN can be treated as a completely opposite architecture. Device-level management is eliminated from networking. Here, the network administrator has the authority to shape the traffic from a centralised control console without configuring individual devices in the network. Unlike traditional networks, where the switch forwards all frames as per a pre-defined program, SDN has the provision to send the decision to switch through an application running on a server. Also, the switch can query the controller to provide information about the current traffic. This results in a highly-scalable, flexible and agile network.

Need for new architecture

There are several reasons why the traditional hardware-centric architecture needs to be replaced.

Increasing number of users. As shown in Fig. 1, advancement in technologies is leading to a drastic increase in the number of Internet users. Technologies and devices are becoming user-friendly. These have now become an inevitable part of our daily lives. This calls for a demand of upgrade in the technology to accommodate the increasing traffic.

Continuously varying traffic pattern. The number of users at any particular place

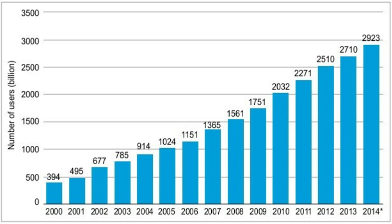

Fig. 1: Increasing number of Internet users (Courtesy: www.statista.com)